

# Kendall Ash

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## EDUCATION

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| <b>Michigan State University</b><br>MS Fisheries & Wildlife; EEB                                     | East Lansing, MI<br>Expected December 2026 |
| <b>University of Michigan</b><br>BS, <i>summa cum laude</i> , Ecology & Evolutionary Biology; German | Ann Arbor, MI<br>May 2024                  |

## FELLOWSHIPS AND AWARDS

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|----------------------------------------------------------------------------------------|-------------|
| John and Marietta Peter Scholarship, College of Agriculture and Natural Resources, MSU | 2025-2026   |
| Robert C. Ball and Betty A. Ball Fisheries and Wildlife Fellowship, MSU                | 2025        |
| International Conference on Harmful Algae Travel Award, NOAA NCCOS                     | 2025        |
| Fisheries and Wildlife GSO Travel Award, MSU                                           | 2025        |
| EEB International Travel Award, MSU                                                    | 2025        |
| Best Prospective Oral Presentation, Fisheries and Wildlife GSO Symposium, MSU          | 2025        |
| James B. Angell Scholar, University of Michigan                                        | 2022 & 2024 |
| University Honors, University of Michigan                                              | 2020-2024   |

## RESEARCH EXPERIENCE

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| Michigan State University<br><i>Graduate Research Assistant in Bridge Lab</i><br><a href="https://bridgelabresearch.org/">https://bridgelabresearch.org/</a> | East Lansing, MI<br>January 2025 – December 2026 |
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Conducting large-scale data analysis on U.S. lakes using the NLA, LAGOS, and CMIP6 databases

- Using machine learning to analyze the drivers of cyanobacteria microcystin production across thousands of lakes in the present day and under future climate change scenarios
- Analyzing cyanobacteria community composition and functional traits using redundancy analysis
- Compiling, processing, and QA/QC'd large amounts of data using RStudio and cloud computing
- Assisting with additional lab projects including analyzing the relationship between lake sampling efforts and community demographics as well as creating the LAGOS LIMNO v.2.0 database

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| University of Michigan Biological Station<br><i>Field Technician</i> | Pellston, MI<br>May 2024 – August 2024 |
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Assisted visiting researchers with projects at the station

- Assembled mesocosm plot for ant colony research with Dr. J. P. Lessard (Concordia University)
- Captured bumble bees for population and pathogen surveys and collected strawberries for chemical analysis with Dr. Mary Jamieson (Oakland University)
- Digitized data archive and herbarium specimens
- Mapped field sites on station property on ArcGIS

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| University of Michigan<br><i>Research Assistant with Dr. Meghan Duffy</i> | Ann Arbor, MI<br>June 2023 – May 2024 |
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Examined relationships and complex interactions between fungal and microsporidian parasites of *Daphnia*

<https://duffylab.wordpress.com/>

- Field work: sampled and analyzed local lakes using YSI multiparameter probe, fluorometer, Secchi disk, and zooplankton nets
- Lab work: assisted with and conducted various research projects, maintained *Daphnia* and parasite colonies, filtered lake water, ran genetic analyses

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| Michigan Medicine<br><i>Research Assistant with Dr. Rosanne Rouf</i> | Ann Arbor, MI<br>August 2021 – January 2022 |
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Used mouse models to examine sex differences in heart failure and cardiovascular disease

- Conducted genetic analysis to identify mice genotypes using DNA extraction, PCR, and gel electrophoresis

## PERSONAL PROJECTS

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### ***Daphnia* feeding assay (Duffy Lab)**

August 2023 – November 2023

Conducted algal feeding assay to compare efficacy of cell quantification methods to assess *Daphnia* feeding

- Quantified algal samples before and after *Daphnia* feeding using ddPCR with and without addition of PMA dye, fluorometry, and cell counting
- Conducted statistical analysis and visualization in RStudio and wrote a comprehensive report assessing efficacy of quantification methods

### **Arctic algal growth experiment (SIT Study Abroad Program)**

April 2023 – May 2023

Ran a temperature-dependent algal growth experiment to project climate change effects on harmful algal blooms in Iceland. Conducted under supervision of Sara Hardardottir at the Marine and Freshwater Research Institute in Hafnarfjörður, Iceland.

- Collected, cultured, and counted *Pseudo-nitzschia* diatoms
- Ran statistical analysis in RStudio, wrote and presented a comprehensive report

## PUBLICATIONS, PRESENTATIONS, AND POSTERS

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Lippert, A.L., Z. Naylor, X. Sun, P.J. Hanly, J. Díaz Vázquez, M. Haite, **K. Ash**, K.S. Cheruvilil, and P.A. Soranno. 2025. User Guide to LAGOS-US HUMAN v1: Data module of human population(1990-2020), urbanization classification, and lake access in the conterminous U.S. ver 1. Environmental Data Initiative.

**K. Ash**. Toxic relationships: predicting drivers of lake microcystin across the U.S. and into the future. Aquatic Confluence: Science, People, Knowledge (ASLO-SIL 2026 Joint Meeting). May 2026.

**K. Ash**. Predicting cyanobacteria blooms in U.S. lakes using national scale data. Oral presentation, International Conference on Harmful Algae, Punta Arenas, Chile. October 2025.

A. Lippert, **K. Ash**. Examining relationships between lake sampling efforts and community demographics. Poster presentation, MSU Ecology, Evolution, and Behavior Symposium. May 2025

**K. Ash**. Predicting occurrences of cyanobacteria harmful algal blooms in U.S. lakes under a changing climate. Oral presentation, MSU Fisheries & Wildlife GSO Symposium. March 2025.

## TEACHING EXPERIENCE

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Michigan State University

Magaliesburg and Sodwana Bay, South Africa

*Teaching Assistant, Undergraduate Internship Experiences in South Africa*

2025

- Coordinated and supported students for two-week cheetah conservation and marine biology internships in South Africa; graded student assignments

Michigan State University

East Lansing, MI

*Grader, Field and Laboratory Techniques for Aquatic Studies*

2025

- Prepared and assisted with laboratory activities and equipment, graded student assignments
- Prepared and presented a guest lecture about algae and harmful algal blooms

## SKILLS

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| • Programming in R, Python, & C++                                  | • Cloud Computing (AWS, MSU HPCC)                                                         |
| • Machine learning (LightGBM, XGBoost, RandomForest)               | • Aquatic field techniques—zooplankton and phytoplankton collection, YSI probes           |
| • Microsoft Office Suite and Adobe Creative Cloud                  | • Insect capture and collection                                                           |
| • RStudio, ArcGIS, QGIS, Google Earth Engine, and Google Earth Pro | • Microscopy, DNA extraction, PCR, ddPCR, gel electrophoresis, cell counting, fluorometry |
| • Climate modeling and remote sensing                              | • Mass spectrometry, chromatography                                                       |
| • Statistical analyses: RDA, PCA, GLM                              | • Open Water SCUBA Certified                                                              |
|                                                                    | • Proficient in German                                                                    |

## AREAS OF INTEREST

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|                       |                                   |                   |                  |
|-----------------------|-----------------------------------|-------------------|------------------|
| Aquatic ecology       | Climate change and Arctic ecology | Microbial ecology | Entomology       |
| Computational ecology | Harmful algal blooms              | Inducible defense | Parasitism       |
|                       |                                   |                   | Biomagnification |

## EXTRACURRICULARS

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|-----------------------------------------------------------|-----------|
| Organizer, iNaturalist City Nature Challenge Lansing      | 2025      |
| Guide at MSU Bug House                                    | 2025      |
| Volunteer at Refugee Development Center                   | 2025      |
| Tutor in biology, chemistry, math, and German             | 2016-2024 |
| University of Michigan Women's Rugby                      | 2023-2024 |
| University of Michigan Entomology Club                    | 2022-2024 |
| Volunteer at Ann Arbor Parks and Recreation & Campus Farm | 2021-2024 |

## RELEVANT COURSEWORK

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|                                    |                                              |
|------------------------------------|----------------------------------------------|
| Introductory and organic chemistry | Protecting & restoring lakes                 |
| General physics I & II             | Evolution                                    |
| Calculus I, II, & III              | Climate modeling & carbon management         |
| Introductory biology               | Arctic: changing ecosystems and resilience   |
| General ecology                    | Fisheries & Wildlife Outreach and Engagement |
| Genetics                           | Applications of Terrestrial Remote Sensing   |
| Biology of insects                 | Statistical Methods in Ecology and Evolution |
| Plant biology                      | Molecular Ecology & Conservation Genetics    |

## REFERENCES

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|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Dr. Kendra Spence<br>Cheruvelil<br>Advisor<br>PI/Co-director<br>Bridge Lab MSU<br>(517) 243-1444<br>ksc@msu.edu | Dr. Patrick Hanly<br>Advisor<br>Project Scientist<br>Bridge Lab MSU<br>(401) 864-7313<br>hanlypat@msu.edu | Dr. Helen Habicht<br>Research coordinator<br>University of<br>Michigan<br>Biological Station<br>(269) 986-2086<br>habicht@umich.edu | Daniel Govoni<br>Study abroad<br>professor/ project<br>mentor<br>School for<br>International Training<br>+354 8468258<br>danielgo@vedur.is |
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